Process Trustworthiness: A Local-First, Tamper-Evident Standard and Reference Implementation for Verifiable Research-Provenance Evidence

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Abstract

Reproducibility has become the dominant frame for scientific integrity: a result is trustworthy insofar as an independent party can rebuild it from shared data and code. We argue that reproducibility verifies *what* was claimed, but not *how* the claim was arrived at, and that the latter is a distinct, complementary pillar of integrity that the current crisis exposes but does not address. We call this pillar *process trustworthiness*: verifiable evidence that the conduct of research—experimentation, reading, writing, and reasoning—was recorded as it happened, under an attested device, and disclosed honestly about its own limits. We motivate this pillar from the literature on the reproducibility crisis, give its scientific rationale and technical feasibility, and present a concrete exemplar: an open, local-first EVIDENCE PACKAGE v1 and its Rust/Tauri reference implementation. The system maintains a tamper-evident hash chain under an ED25519 device signer, encrypts sensitive content with XCHACHA20-POLY1305, canonicalizes records with RFC 8785 JCS, and exports a two-layer package that any third party can verify *offline*. Crucially, the design refuses to certify identity, authorship, originality, or completeness: it discloses capability boundaries honestly (including platforms where capture is *unavailable*) and records its own gaps. We close with the scope, limitations, and a governance path for adopting process-provenance evidence as a community standard.

1 Introduction

The reproducibility crisis is frequently told as a story about *results*: too many published findings cannot be rebuilt from the materials authors share [Nosek et al., 2015, Stodden, 2009]. The standard remedy is therefore to share more—data, code, protocols, preregistrations—so that the result can be re-derived [Nosek et al., 2015, Stodden, 2009]. This frames scientific integrity almost entirely around *reproducibility*: the capacity of an independent actor to regenerate a claimed outcome.

We propose a complementary frame. Reproducibility answers the question *"can the result be rebuilt?"* It does not answer *"was the result arrived at honestly?"* A finding can be perfectly reproducible yet the product of selective reporting, post-hoc hypothesizing, undisclosed reuse of a collaborator's work, or an AI-assisted draft presented as original. Conversely, a genuinely honest workflow may fail to reproduce for reasons unrelated to misconduct—fragile materials, unreported environmental conditions, or legitimate stochasticity [Ioannidis, 2005]. The two pillars are correlated but not the

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Code and specification: `https://github.com/chongliuresearch/academic-integrity-recorder`.

same, and the crisis literature documents the blind spot explicitly: even when studies are replicated, scholars cannot tell whether the *process* that produced the original was faithful [John et al., 2012, Fanelli, 2009].

We call the missing pillar **process trustworthiness**. It is the property that the conduct of research was recorded as it occurred, bound to an attested device, and accompanied by an honest account of what was and was not observed. Process trustworthiness is *not* a claim of integrity, originality, or identity. It is a verifiable record of evidence about process, with its limitations made part of the record.

Our contributions are:

1. a clear separation of *reproducibility* (what) from *process trustworthiness* (how), with an argument that the latter is a distinct scientific-integrity pillar;

2. a design for local-first, tamper-evident, privacy-preserving, and *honestly bounded* process evidence, including an Evidence Package v1 specification;

3. an open reference implementation (Academic Integrity Recorder) in Rust (core), Tauri (desktop), and browser/VS Code/shell extensions, with an offline verifier;

4. a candid treatment of scope, adversarial limits, and a governance path for adoption.

> **Non-claim.** A process-evidence package proves byte integrity and a device signature. It does *not* certify identity, authorship, originality, or academic integrity, and it cannot prove that all research activity was captured. This boundary is enforced in code and in the exported manifest, not merely asserted in prose.

2 The Reproducibility Crisis

2.1 What the empirical record shows

The crisis is not anecdotal. In a 2016 *Nature* survey of 1,576 researchers, **70%** reported having failed to reproduce another group's experiment, and more than half reported failing to reproduce *their own* [Baker, 2016]. Among large coordinated replications, the Open Science Collaboration reran 100 psychology studies and found only **36%** reproduced the original statistically significant result, with effect sizes roughly half as large [Open Science Collaboration, 2015]. In preclinical cancer biology, the Reproducibility Project reproduced only a minority of high-impact findings [Errington et al., 2021], consistent with earlier industry reports that roughly **11%** (6 of 53) of landmark preclinical studies could be reproduced internally [Begley and Ellis, 2012] and that a major pharmaceutical pipeline observed reproducibility in only about one fifth of projects [Prinz et al., 2011].

These numbers are not a verdict on any discipline's honesty; they are a verdict on the *visibility* of process. As Ioannidis [2005] argued two decades ago, the prevailing incentive structure predicts a large fraction of false positives even without malfeasance, and John et al. [2012] documented that questionable research practices (p-hacking, selective reporting, undisclosed flexibility) are common and often unconscious.

2.2 Why reproducibility alone is insufficient

Reproducibility operates on the *output* side of research. It asks whether, given shared inputs, the same outputs reappear. Three gaps follow.

First, **the reporting gap**. Reproducibility depends on what authors choose to share. Studies show extensive under-disclosure of experimental design, and that the absence of a disclosure norm is itself a cause of non-reproduction [Nosek et al., 2015]. What is never written down cannot be reproduced, and the decision of what to write down is exactly the process reproducibility cannot see.

Second, **the selectivity gap**. Even with full data, reproducibility cannot distinguish a result obtained by pre-specified analysis from one obtained by exploring many analyses and reporting the best [Ioannidis, 2005, Gelman and Loken, 2014]. Preregistration addresses this prospectively [Nosek et al., 2018], but it covers a narrow slice of ordinary research behavior and is silent about the long, messy trail of reading, drafting, and revision that precedes any registration.

Third, **the provenance gap**. Modern research increasingly involves tools—search, writing environments, version control, and AI assistants—whose use is rarely disclosed in a verifiable way [Gao et al., 2023]. Whether and how an AI contributed to a manuscript, whether a passage was copied from a collaborator, or whether a figure was regenerated after peer review are questions about *process*, and reproducibility is structurally blind to them.

None of this diminishes reproducibility; it motivates a second pillar that observes the conduct of research directly rather than inferring it from its outputs.

3 Process Trustworthiness: Rationale and Feasibility

3.1 Scientific rationale

Process trustworthiness is continuous with long-standing ideas in the philosophy and practice of science. The laboratory notebook is a centuries-old technology for exactly this purpose: a contemporaneous, append-only record that makes the conduct of experiment visible to later scrutiny [Macrina, 2017]. The W3C PROV data model formalizes provenance as a first-class, machine-checkable entity [World Wide Web Consortium, 2013, Moreau and Missier, 2013, Moreau and Groth, 2015], and open-science movements have long argued that transparency of process is as important as availability of products [Nosek et al., 2015, Fanelli, 2011].

Epistemically, the case is simple. A scientific claim is warranted not only by its being re-derivable but by the trustworthiness of the pathway that produced it. When that pathway is recorded *as it happened* and *bound to an attested device*, a later reader gains a new kind of evidence: not that the result is true, but that the reported process is the actual one. This is weaker than proof and stronger than assertion. It is, in effect, the same role a version-control history or a signed build plays in software engineering—except applied to the reasoning itself.

Process trustworthiness is also *complementary*, not competitive, with reproducibility. The two can be combined: share the data and code (reproducibility) *and* attach a verifiable process record (trustworthiness). Where reproducibility fails for innocent reasons, the process record may still stand; where reproducibility succeeds but the process was deceptive, the record is what reveals it.

3.2 Technical feasibility

Three properties make process evidence practical today where it was not a decade ago.

Local-first, privacy-preserving capture. Evidence can be collected on the researcher's own machine without a server, without surveillance, and with sensitive content never leaving the device in cleartext. This removes the principal objection to process monitoring—that it is a panopticon—and aligns with data-minimization norms [Voigt and Von dem Bussche, 2017].

Cheap, standard cryptography. A tamper-evident record is a hash chain: each event commits to its predecessor, so any deletion, reordering, or edit breaks the chain [Crosby and Wallach, 2009]. A device signature (Ed25519) attests the recorder; canonicalization (RFC 8785 JCS) makes hashing serialization-independent [Rundgren, 2020]; authenticated encryption (XChaCha20-Poly1305) protects sensitive content; and an internet-independent timestamp (e.g., OpenTimestamps) can later prove existence without trusting a notary [OpenTimestamps, 2018]. None of these are exotic; all are library-grade.

Offline, adversarial verification. Because integrity is mathematical, a skeptic needs no special access to check a package. The verifier is a small, auditable program that re-derives every hash and signature. This is the same principle that makes signed commits and checksums trustworthy without a trusted third party.

4 A Proposed Exemplar: Evidence Package v1 and Reference Implementation

We present Academic Integrity Recorder, an open-source implementation of the above, and the Evidence Package v1 it emits. The design obeys five principles: *local-first*, *tamper-evidence*, *honest capability disclosure*, *privacy-by-default*, and *offline verification*.

4.1 Architecture

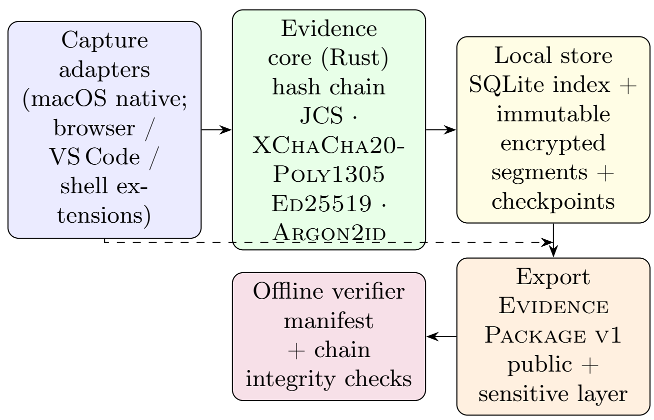

Figure 1: Local-first architecture. Evidence flows from platform capture adapters into a Rust evidence core that maintains a tamper-evident hash chain under an Ed25519 device signer, is persisted in an immutable local store, exported as a two-layer Evidence Package v1, and verified entirely offline. No telemetry leaves the device.

Capture adapters observe activity—foreground application, file events, browser and editor interactions, shell commands (opt-in), and screenshots of the selected window on supported platforms. They forward events to the evidence core, which canonicalizes, hashes, signs, and persists them (Figure 1).

4.2 The evidence event and hash chain

Each recorded moment is an *evidence event* with a monotonic sequence number, occurrence and capture timestamps, a source, a semantic kind (e.g., input, file-modified, web-interaction, annotation, anchor-created, AI-disclosure), a sensitivity label, and a payload. The core computes

$$\begin{aligned} h_{\text{payload}} &= \text{SHA256}(\text{RFC 8785 JCS}(\text{payload})), \\ h_{\text{event}} &= \text{SHA256}(\text{RFC 8785 JCS}(\text{material})), \end{aligned}$$

where material binds the event's identity, project, timestamps, source, kind, sensitivity, h_{payload}, and the *previous* event hash h_{prev}. The first event's predecessor is a fixed 64-zero genesis hash. The result is a linked, tamper-evident chain: altering, deleting, or reordering any event changes its h_{event} and thus breaks every subsequent link and the signed head [Crosby and Wallach, 2009].

Canonicalization uses RFC 8785 JCS (UTF-16 property-name ordering, I-JSON safe-integer range) so that hashing is independent of key order or whitespace [Rundgren, 2020]. The device attests each signed high-water checkpoint with Ed25519 [Bernstein et al., 2012]; the fingerprint published in the manifest is SHA256 of the public key, and sensitive content is sealed with XChaCha20-Poly1305 [Bernstein, 2008].

4.3 Storage with crash recovery

Events are persisted as immutable, create-only encrypted `.segment` files named by zero-padded sequence, alongside an encrypted `objects/` store for captured artifacts and a `checkpoints/` directory of signed high-water markers (`signed-high-water/v1`). Each append writes the segment (with `fsync`), writes a *signed pending-append* record, then indexes and checkpoints, then removes the pending record. On open, the store reconciles any surviving pending append and rejects unsigned extra segments, missing or reordered segments, and truncation of the signed tail [Crosby and Wallach, 2009]. Legacy stores require an explicit, honestly bounded migration that seals validated local bytes without claiming to prove earlier existence.

4.4 Two-layer export and honest disclosure

Export produces a single Evidence Package v1 (a ZIP) with a signed manifest. The *public layer* contains `evidence.json` (events with payloads replaced by their hashes), a bilingual `report.html` (zh-CN/en), an English-summary `report.pdf`, and a `timestamp-target.json` for optional OpenTimestamps. The *sensitive layer* holds the full events, research items, artifacts, anchors, and AI disclosures, encrypted under a passphrase derived with Argon2id (Argon2id) [Biryukov et al., 2015]. The manifest carries an explicit `evidenceClaim` and a `limitations` list, and embeds a `capabilityReport` recording, per platform capability, one of Available, PermissionRequired, Degraded, or Unavailable.

This last point is the design's distinctive honesty mechanism. The macOS adapter is implemented; on Windows and Linux the same adapter reports capture as Unavailable rather than fabricating capability [Academic Integrity Recorder Project, 2026c]. Gaps are first-class evidence: explicit `Gap`/`Redaction` events record *why* coverage lapsed (permission denied, platform limitation, adapter failure, user pause/exclusion, content deleted/redacted, data unavailable). Manuscript anchors bind a research claim to a quote in a PDF/DOCX/TeX/Markdown source by SHA256 of the quote and surrounding context, and can be revalidated as Valid, Relocatable, or Stale [Academic Integrity Recorder Project, 2026b]. An *active-time* algorithm reports unique observed foreground activity, capped by a timeout and reset on pause, lock, sleep, or backgrounding—a conservative lower bound, never an over-claim.

4.5 Offline verification

A skeptic runs the verifier on the package alone. It checks the manifest Ed25519 signature, the final checkpoint signature, continuity of the public hash chain, every file's size and SHA256 digest against the manifest, and—given the sensitive-layer passphrase—that payload hashes reproduce, the event chain is complete, active time is consistent, and declared relationships hold. It rejects duplicate or undeclared ZIP entries, digest mismatches, and wrong passphrases. Success proves *integrity and a device signature only*; the report states this in its output.

5 Scope, Limitations, Governance, and Adoption

5.1 Scope and what the record can show

Process trustworthiness is most valuable where the contested question is *how*, not *whether*: disclosure of AI assistance, provenance of a passage, the trail of revision, the existence of a planned experiment that was later dropped, or a contemporaneous lab record. It strengthens theses, grant reports, and post-publication audit alike.

5.2 Honest limitations and adversarial considerations

The system makes no claim to completeness or proof of integrity, and the verifier says so. Specifically:

- **Coverage upper bound.** Only observed activity is recorded. Offline work, use of an uninstrumented machine, or delegation to a colleague creates gaps the record cannot fill, and the design records these as gaps rather than hiding them [Academic Integrity Recorder Project, 2026d].
- **Platform dependence.** Capture fidelity is bounded by what the OS permits; where an adapter is UNAVAILABLE, the record says so [Academic Integrity Recorder Project, 2026a].
- **Not identity.** A device signature attests a key, not a legal person. It cannot prevent one person from using another's device or account.
- **Adversarial bypass.** A determined adversary can record on a separate machine, disable capture, or fabricate inputs before recording; the system raises the cost and leaves traces (gaps, inconsistencies) but is not a tamper-proof oracle.
- **Not a substitute for peer review.** The record is evidence to weigh, not a verdict on merit or originality.

These limits are features when stated plainly: a process record is trustworthy *because* it is honest about where it stops.

5.3 Governance and adoption path

We propose process-provenance evidence as a lightweight, open community standard, in the spirit of how code signing and transparency logs became default expectations [Crosby and Wallach, 2009]. Concrete steps:

1. **Open specification and interop.** The EVIDENCE PACKAGE v1 format and verifier are open; journals or archives can run the verifier on deposited packages without vendor lock-in [Academic Integrity Recorder Project, 2026b,c].
2. **Complement, don't replace.** Tie the record to existing norms: the data-availability statement, the preregistration, and the conflicts/disclosure form. The AI-disclosure entity maps directly onto emerging AI-use policies [Gao et al., 2023, Committee on Publication Ethics (COPE), 2023].
3. **Tiered adoption.** Start with self-archiving and reviewer-on-request, then optional journal encouragement, then (where appropriate) a deposition requirement for certain claim types. Voluntary, low-friction adoption avoids the surveillance objection.

4. **Incentives for honesty.** Because gaps are recorded, a complete, gap-light record becomes a positive signal; a record full of disclosed gaps is at least honest, which is more than the current default of silence.

6 Conclusion

Reproducibility verifies *what* was claimed; verifiable process provenance verifies *how* the claim was arrived at. We have argued that the second is a distinct, complementary pillar of scientific integrity, grounded in the epistemics of the laboratory record and made practical by local-first, cryptographic, offline verifiable evidence. Academic Integrity Recorder and the Evidence Package v1 show that such evidence can be collected without surveillance, encrypted by default, and verified by anyone—while refusing to certify more than it can prove. Adopting process trustworthiness will not end the reproducibility crisis, but it gives the scientific community a new, honest instrument for the part of integrity that reproducibility was never designed to see.

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